

Design Thinking Lab 1 Report

Topic: Electricity Outage Information Is Not Communicated Clearly

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Step 1: Observation

Observation Log

Context observed:

Unplanned and planned electricity outages in residential and semi-urban areas; information accessed via electricity board offices, helpline numbers, SMS alerts, and social media.

User Behaviour

- Users repeatedly check power switches and meters to confirm outage.
- Many users call customer care numbers multiple times.
- Some users depend on neighbors or local electricians for information.
- Businesses switch to generators without knowing outage duration.

Delays, Confusion, and Frustration

- No prior notification for sudden outages.
- Helpline numbers are busy or provide generic responses.
- Conflicting information from different sources.
- Users are unsure about restoration time.

Environmental / System Constraints

- Limited mobile network during outages.
- No centralized, real-time communication platform.
- Manual updates by field staff cause delays.

Documentation

- Notes from local observations.
- Screenshots of unanswered helpline calls.
- Photos of notice boards with outdated outage information.

Step 2: User Identification

Stakeholder List

User Group	Role	Expectations
Residential Consumers	Daily electricity users	Timely and accurate outage updates
Electricity Board Staff	Maintenance and fault repair	Reduced complaint calls and clear communication
Commercial Users (shops, hospitals)	Depend on continuous power	Advance notice and restoration timelines
Local Administrators	Information coordination	Fewer public complaints and transparency

Output: Stakeholder Identification

Step 3: Interviews / Surveys

Interview Questionnaire (Open-ended)

- How do you usually find out about a power outage?
- What information do you expect during an outage?
- Have you ever received incorrect or delayed outage information?
- How does poor communication affect your daily routine?
- Which communication method do you trust the most?
- What improvements would you suggest?
- How often do outages occur in your area?

Interview Summary (30 Users)

Timestamp	Name of the person	What is the main reason rural citizens struggle to	What is one major effect of poor digital access in	Which of the following is a digital public service?	Digital literacy reasons	Which factor most affects internet access in rural	Which group is most affected by the digital divide	Which solution can help rural citizens access digi	What does "digital divide" mean?
1/25/2024 13:09:48	Uppala Sheerane	High urban population	Delay in receiving government services	Online ration card application	Lack of skills to use digital d	Too many mobile towers	Farmers and elderly people	Digital literacy programs	Gap between rich and poor
1/25/2024 13:04:28	Syed	Too many smartphones	Delay in receiving government services	Online ration card application	Lack of skills to use digital d	High population	IT workers	Digital literacy programs	Gap between those with and without i
1/25/2024 13:07:29	Silbilatha	Poor internet connectivity	Delay in receiving government services	Online ration card application	Lack of skills to use digital d	Poor network infrastructure	IT workers	Digital literacy programs	Gap between those with and without i
1/25/2024 13:12:39	Sai Akshay	Poor internet connectivity	Delay in receiving government services	Online ration card application	Lack of skills to use digital d	Poor network infrastructure	Farmers and elderly people	Digital literacy programs	Gap between those with and without i
1/25/2024 14:44:20	Nagalakshmi	Poor internet connectivity	Delay in receiving government services	Online ration card application	Lack of skills to use digital d	Poor network infrastructure	Farmers and elderly people	Digital literacy programs	Gap between those with and without i
1/21/2024 8:34:26	Shahk Mohammad Abdul I	Too many smartphones	Delay in receiving government services	Paper voting slip	Lack of schools	Lack of roads	IT workers	Digital literacy programs	Gap between rich and poor
1/21/2024 8:35:14	Yaseen	Excessive online content	Faster communication	Post office letter	Not knowing how to read box	Poor network infrastructure	Farmers and elderly people	Free internet cafes	Gap between rich and poor
1/21/2024 8:39:24									
1/21/2024 8:37:23	Kardik	Poor internet connectivity	Better education	Online ration card application	Lack of schools	High population	School teachers	Free internet cafes	Difference between schools
1/21/2024 8:47:47	Shahk Sheema Tabassum	Excessive online content	Better education	Online ration card application	Lack of skills to use digital d	Poor network infrastructure	Farmers and elderly people	Digital literacy programs	Gap between those with and without i
1/21/2024 9:18:59	Syed Murtuza Khadi	Poor internet connectivity	Delay in receiving government services	Post office letter	Lack of skills to use digital d	Lack of roads	School teachers	Free internet cafes	Gap between rich and poor
1/21/2024 9:20:27	M.ameen	Poor internet connectivity	Delay in receiving government services	Online ration card application	Lack of skills to use digital d	Poor network infrastructure	Farmers and elderly people	Digital literacy programs	Gap between those with and without i
1/21/2024 11:54:35	S. ABU SAKKAR	Poor internet connectivity	Delay in receiving government services	Online ration card application	Lack of skills to use digital d	Poor network infrastructure	Farmers and elderly people	Digital literacy programs	Gap between those with and without i
1/21/2024 12:05:32	Shamshya	Poor internet connectivity	Delay in receiving government services	Online ration card application	Lack of skills to use digital d	Poor network infrastructure	IT workers	Digital literacy programs	Gap between those with and without i
1/22/2024 13:35:39	Tabassum	Poor internet connectivity	Easy banking	Online ration card application	Lack of skills to use digital d	Poor network infrastructure	Farmers and elderly people	Digital literacy programs	Gap between those with and without i
1/22/2024 13:53:55		Excessive online content	Better education	Online ration card application	Lack of skills to use digital d	Too many mobile towers	Urban professionals	Free internet cafes	Difference between schools
1/22/2024 13:56:16	Syed. Maha jabeen	Excessive online content	Better education	Online ration card application	Lack of skills to use digital d	Too many mobile towers	Farmers and elderly people	Free internet cafes	Difference between schools
1/22/2024 14:43:24	TM Raghava	Poor internet connectivity	Delay in receiving government services	Online ration card application	Lack of skills to use digital d	Poor network infrastructure	Farmers and elderly people	Digital literacy programs	Gap between those with and without i
1/22/2024 14:47:30	Chiru	Too many smartphones	Faster communication	Post office letter	Not knowing how to read box	High population	School teachers	Free internet cafes	Gap between those with and without i
1/22/2024 15:21:29	Sharmadha Vardhan	Poor internet connectivity	Delay in receiving government services	Online ration card application	Lack of skills to use digital d	Poor network infrastructure	IT workers	Digital literacy programs	Gap between those with and without i
1/22/2024 15:27:15	Disha	Too many smartphones	Delay in receiving government services	Post office letter	Not knowing how to read box	Too many mobile towers	Farmers and elderly people	More paperwork	Gap between rich and poor
1/22/2024 15:02:49	Arnav	Poor internet connectivity	Delay in receiving government services	Online ration card application	Lack of skills to use digital d	Poor network infrastructure	Farmers and elderly people	Digital literacy programs	Gap between those with and without i
1/22/2024 15:37:53	Raghu Pulmelra	Poor internet connectivity	Delay in receiving government services	Online ration card application	Lack of skills to use digital d	Poor network infrastructure	Farmers and elderly people	Digital literacy programs	Gap between those with and without i
1/22/2024 17:43:35		Too many smartphones	Better education	Paper voting slip	Not having electricity	Poor network infrastructure	Farmers and elderly people	Closing village offices	Difference between cities
1/22/2024 20:30:15	A. Nitin	Too many smartphones	Delay in receiving government services	Paper voting slip	Not having electricity	Too many mobile towers	Farmers and elderly people	Digital literacy programs	Gap between rich and poor
2/3/2024 13:01:49	Rajesh	Poor internet connectivity	Delay in receiving government services	Online ration card application	Not knowing how to read box	Poor network infrastructure	School teachers	Free internet cafes	Gap between those with and without i
3/3/2024 13:02:18	Disha	Excessive online content	Delay in receiving government services	Online ration card application	Lack of schools	Poor network infrastructure	Farmers and elderly people	Free internet cafes	Difference between cities
3/3/2024 13:02:32	Shravan	Too many smartphones	Faster communication	Village meeting	Lack of schools	Lack of roads	School teachers	Free internet cafes	Gap between those with and without i
3/3/2024 13:04:02	M.Aseem Reddy	Poor internet connectivity	Easy banking	Village meeting	Lack of skills to use digital d	High population	School teachers	Digital literacy programs	Gap between those with and without i
3/3/2024 13:04:02	Jayash	Poor internet connectivity	Delay in receiving government services	Online ration card application	Not knowing how to read box	High population	Farmers and elderly people	Closing village offices	Gap between rich and poor

Common User Statements

- “We never know how long the power will be gone.”
- “Customer care never answers during outages.”
- “If we knew the timing, we could plan better.”

- “Messages come after power is restored.”
- “Hospitals and shops suffer huge losses.”

Step 4: Pain-Point Analysis

Pain-Point Table

Pain Point	Category
No advance notification	Functional
Unclear restoration time	Functional
Anxiety and stress	Emotional
Loss of work and income	Emotional
No single reliable information source	Systemic
Poor coordination between departments	Systemic

Most Critical Pain Point:

Lack of clear and timely outage communication

Step 5: Root Cause Identification

5-Why Analysis

Problem: Users do not receive clear outage information.

1. Why? → Updates are delayed or missing.
2. Why? → Information depends on manual reporting.
3. Why? → Field staff lack real-time update tools.
4. Why? → No centralized digital communication system.
5. Why? → Low investment in customer communication infrastructure.

Root Cause:

Absence of an integrated, real-time outage communication system

Step 6: Wicked Problem Understanding

Wicked Problem Explanation

- **Difficult to define clearly:**
Communication failure varies by region, outage type, and user dependency.

- **Multiple stakeholders:**
Consumers, electricity boards, administrators, emergency services.
- **Conflicting needs:**
Cost efficiency vs. real-time transparency.
- **No single solution:**
Requires technology, policy changes, training, and coordination.

Problem Classification: Wicked Problem

Step 7: Reflection

1. What assumptions were incorrect?

Initially, it was assumed that **electricity providers already inform users properly** and that people can easily **find outage details online or through SMS**. However, user feedback showed that **notifications are delayed, unclear, or sometimes not received at all**, and many users **do not know the restoration time or reason for the outage**.

2. How did user perspectives change your understanding?

User perspectives revealed that the problem is not only technical but also **communication-related**.

People feel **frustrated, unprepared, and anxious** during power cuts because they lack **clear, timely, and reliable information**.

This shifted the understanding from a simple outage issue to a **user experience and information transparency problem**.

3. Why is jumping to solutions risky?

Jumping directly to solutions (like creating an app or sending SMS alerts) without understanding users may result in **tools that people cannot access, trust, or use effectively**. Proper research is needed to ensure the solution provides **real-time, simple, and widely reachable communication** that truly solves the users' problem.

4. What skills did you develop through this lab?

This lab helped develop **empathy for users, problem observation, user research, critical thinking, and solution planning skills**.

It also improved the ability to **analyze real-world communication gaps and design user-centered digital solutions**.

